

MACQUARIE University
SYDNEY·AUSTRALIA

Advanced Machine Learning - COMP8221

Unit Guide

Dr. Qiongkai Xu, qiongkai.xu@mq.edu.au, School of Computing

Welcome to COMP8221!

Welcome to this unit!

The teaching team:

- Convenor/Lecturer: Qiongkai Xu, qiongkai.xu@mq.edu.au (week 1 - week 6, week 13)
- Lecturer: Zitai Qu, zitai.qu@mq.edu.au (week 7 - week 13)
- Tutor: Utsav Maskey, utsav.maskey@mq.edu.au
- Tutor: Alireza Yegane, alireza.yegane@mq.edu.au

Unit Schedule

Table 1: Unit Schedule

Week	Topic
1-4	Deep generative models (Graphical models, VAE, Diffusion models, etc.)
5-8	Graph neural networks
9-12	Deep reinforcement learning
13	Revision

Assessment Tasks

Table 2: Assessment Tasks

Name	Weighting	Type	Due
Assignment 1	30%	Project	20/09/2026
Assignment 2	30%	Project	01/11/2026
Exam	40%	Examination	Exam period

- Assignments to be submitted through iLearn.

Delivery and Recources

- Classes: each week has two hours of lectures and two hours of workshops.
- Recommended Texts: will be provided during the lecturing.
- Unit web page: will be hosted in iLearn. You will need to log in to iLearn using you student one ID and password. Please post questions in the discussion boards, the teaching team will monitor it.
- Methods of communication: via your university email or through announcements in iLearn.
- Technology used: will make use of a range of modules that are available via the Anaconda Python distribution.

Class time/venue

- Lecturing: 1:00 PM - 3:00 PM, Monday, 14SCO T3 Lecture Theatre
- Choose one of the four Workshops (**start from week 2**):
  - 3:00 PM - 5:00 PM, Monday, 04RPD G14 Computer Lab
  - 5:00 PM - 7:00 PM, Monday, 04RPD G14 Computer Lab
  - 9:00 AM - 11:00 AM, Tuesday, 04RPD G14 Computer Lab
  - 11:00 AM - 1:00 PM, Tuesday, 04RPD G14 Computer Lab

Requirement

- Requirement to Pass this Unit To pass this unit, you must achieve a total mark equal to or greater than 50%.
- Late Assessment Submission Penalty Unless a Special Consideration request has been submitted and approved, a 5% penalty (of the total possible mark of the task) will be applied for each day a written report or presentation assessment is not submitted, up until the 7th day (including weekends). After the 7th day, a grade of '0' will be awarded even if the assessment is submitted. The submission time for all uploaded assessments is 11:55 pm. A 1-hour grace period will be provided to students who experience a technical concern.
- For any late submission of time-sensitive tasks, such as scheduled tests/exams, performance assessments/presentations, and/or scheduled practical assessments/labs, please apply for Special Consideration.

Suggestions

Contact Our Team:

- Questions to lectures: email lecturers during their weeks or ask them after/during lectures
- Questions to workshops: email your TA or ask them after/during workshops
- General Questions: iLearn (General discussion Forum)
- Private Questions: email / iLearn (Privacy Message to Unit Contacts)

Dr Xu's Office Hour: Wednesday 12-1 pm, in Room 357, 4RPD, by appointment.

Suggestions

MACQUARIE University

To approach possibly one of the most theoretically & technically challenging units in computing,

- Attend lectures and workshops.
- Try to understand the content as deeply as possible by
  - deriving the theory
  - programming the process
  - evaluating & analysing the outcomes.
  - ...
- Ask (good) questions.
- Be critical.